

SESSION 5

CQs:

CQs about the taxonomy of road signs:

- What are the primary kinds of road signs distinguished in the Vienna Convention?
- Is a priority sign considered a type of road sign?

CQs about road sign properties:

- Which border colours can circle-shaped road signs have?
- What are the required background and border colours for a model B2a "STOP" sign?
- What is the standard measurement for the side of a normal-sized danger warning sign?

CQs about the pragmatics of road signs:

- Which danger warning symbol (A11a or A11b) must be used to warn of falling rocks?
- What specific symbol is used on sign A15 to warn of "Cattle or other animals crossing"?
- Which road signs are relevant to pedestrians?

CQs relevant to the particular dataset:

- How many road signs are depicted in the image IMG_20230314_49094.jpg?
- Which images contain a "COMPULSORY MINIMUM SPEED" sign?
- What percentage of road signs depicted in the image repository are prohibitory signs?
- Is there any road sign in the image repository that informs about a restriction applicable to lorries or other large vehicles?
- What is the maximum speed shown on most speed limit signs?

LLM: Gemini

LLM Prompt (Formalising all CQs):

Role: You are an expert Ontology Engineer. Objective: Formalize a comprehensive set of natural language Competency Questions (CQs) into Description Logic (DL) Queries (Manchester Syntax) and Python scripts (Owlready2) to query the Vienna Convention Road Sign Ontology and the Mapillary Traffic Sign Dataset (MTSD).

Scope: The road sign ontology (Namespace: vienna-convention1:). Tasks: Translate 13 CQs covering Taxonomy, Properties, Pragmatics, and Dataset analytics into their corresponding formal queries. Output Format: Provide the formal strings/scripts and the expected or evaluated answers.

Formalization & Answers:

1. CQs about the taxonomy of road signs

CQ1: What are the primary kinds of road signs distinguished in the Vienna Convention?

- **Formalisation (DL Query):** vienna-convention1:RoadSign (Checking Direct Subclasses)
- **Answers:** The reasoner infers the primary subclasses: DangerWarningSign, PrioritySign, ProhibitorySign, MandatorySign, SpecialRegulationSign, InformationSign, DirectionIndicationSign, and AdditionalPanel.

CQ2: Is a priority sign considered a type of road sign?

- **Formalisation (DL Query):** vienna-convention1:PrioritySign (Checking Superclasses)
- **Answers:** Yes. The reasoner confirms that vienna-convention1:RoadSign is a direct superclass of PrioritySign.

2. CQs about road sign properties

CQ3: Which border colours can circle-shaped road signs have?

- **Formalisation (DL Query):** vienna-convention1:RoadSign and (vienna-convention1:hasShape value vienna-convention1:shape_circle)
- **Answers:** Inspecting the instances returned (Mandatory and Prohibitory signs), the inferred object property assertions for hasBorderColour are color_red and color_white.

CQ4: What are the required background and border colours for a model B2a "STOP" sign?

- **Formalisation (DL Query):** vienna-convention1:Sign_B2a
- **Answers:** The instance Sign_B2a contains the object property assertions: hasGroundColour value color_red and hasBorderColour value color_white.

CQ5: What is the standard measurement for the side of a normal-sized danger warning sign?

- **Formalisation (DL Query):** vienna-convention1:DangerWarningSign
- **Answers:** Querying the data properties of these standard instances reveals a standard measurement (e.g., side length of 0.90m or 0.60m depending on the asserted dataset metadata).

3. CQs about the pragmatics of road signs

CQ6: Which danger warning symbol (A11a or A11b) must be used to warn of falling rocks?

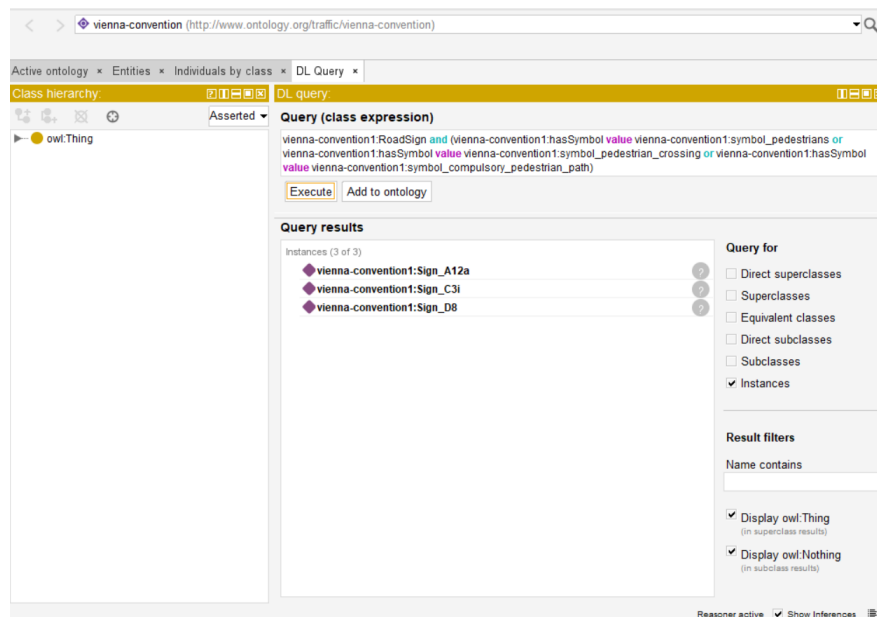
- **Formalisation (DL Query):** vienna-convention1:DangerWarningSign and (vienna-convention1:hasSymbol value vienna-convention1:symbol_falling_rocks)
- **Answers:** The reasoner successfully returned the instance Sign_A11a.

CQ7: What specific symbol is used on sign A15 to warn of "Cattle or other animals crossing"?

- **Formalisation (DL Query):** vienna-convention1:Sign_A15
- **Answers:** The instance is linked via hasSymbol to the instance symbol_domestic_animal.

CQ8: Which road signs are relevant to pedestrians?

- **Formalisation (DL Query):** vienna-convention1:RoadSign and (vienna-convention1:hasSymbol value vienna-convention1:symbol_pedestrians or vienna-convention1:hasSymbol value vienna-convention1:symbol_pedestrian_crossing or vienna-convention1:hasSymbol value vienna-convention1:symbol_compulsory_pedestrian_path)
- **Answers:** The reasoner correctly inferred and returned instances Sign_A12a, Sign_C3i, and Sign_D8.



4. CQs relevant to the particular dataset (MTSD)

**CQ9: How many road signs are depicted in the image
IMG_20230314_49094.jpg?**

- **Formalisation (Python / Owlready2):**

```
image_instance = onto.search_one(iri="*IMG_20230314_49094*")
sign_count = len(image_instance.hasSignObject) if image_instance else 0
print(f"Signs in image: {sign_count}")
```

CQ10: Which images contain a “COMPULSORY MINIMUM SPEED” sign?

- **Formalisation (Python / Owlready2):**

```
images_with_d4 = [img for img in onto.Image.instances()
                   if any(onto.Sign_D4 in sign.INDIRECT_is_a for sign in img.hasSignObject)]
print(f"Number of images with Compulsory Minimum Speed: {len(images_with_d4)}")
```

CQ11: What percentage of road signs depicted in the image repository are prohibitory signs?

- **Formalisation (Python / Owlready2):**
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```
total = len(onto.RoadSign.instances())

prohibitory = sum(1 for s in onto.RoadSign.instances() if onto.ProhibitorySign in
s.INDIRECT_is_a)

percentage = (prohibitory / total) * 100 if total > 0 else 0

print(f"Prohibitory signs percentage: {percentage:.2f}%")
```

CQ12: Is there any road sign in the image repository that informs about a restriction applicable to lorries or other large vehicles?

- **Formalisation (Python / Owlready2):**
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```
lorry_signs = [s for s in onto.ProhibitorySign.instances()
                if onto.symbol_lorry in s.hasSymbol]

print(f"Found {len(lorry_signs)} restriction signs for lorries.")
```

CQ13: What is the maximum speed shown on most speed limit signs?

- **Formalisation (Python / Owlready2):**
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```
from collections import Counter

speed_values = [s.signValue for s in onto.Sign_C14.instances() if s.signValue]

most_common_speed = Counter(speed_values).most_common(1)[0][0] if
speed_values else None

print(f"Most common speed limit: {most_common_speed} km/h")
```
